A Finite Counterexample Challenge for Equation 677

May 2026

Problem 1 (Counterexample question)**.** Assume that the finite implication

$$\text{E677} \models_{\text{fin}} \text{E255}$$

is false. Find a finite nonempty magma $(M, \diamond)$ satisfying

$$\text{E677}: \qquad x = y \diamond \big(x \diamond ((y \diamond x) \diamond y)\big)$$

for all $x, y \in M$, together with a point $x \in M$ for which

$$\text{E255}: \qquad x = ((x \diamond x) \diamond x) \diamond x$$

fails. Prefer a smallest example, or a branch analysis that narrows what the smallest example must look like.

This is the companion problem to the proof challenge. Here the working attitude is the opposite: suppose a bad finite point exists, then force its local diagram until either a complete Cayley table appears or the branch collapses.

Warning. The identities below are necessary conditions for a counterexample, not a substitute for checking the full law E677. Do not assume associativity, commutativity, an identity element, inverses, right cancellation, or quasigroup structure. Every cancellation below is left cancellation, justified by finite left division.

Hints

1. **Start with the universal tools.** In any finite E677-magma, every left translation $L_a(t) = a \diamond t$ is bijective, and

$$a \backslash b = b \diamond ((a \diamond b) \diamond a). \tag{L}$$

Thus left cancellation is available. Also keep the two global identities

$$z = (y \diamond z) \diamond \big((y \diamond (y \diamond z)) \diamond y\big), \tag{T}$$

$$(y \diamond (y \diamond z)) \diamond y = z \diamond \big(((y \diamond z) \diamond z) \diamond (y \diamond z)\big). \tag{K}$$

These are the main algebraic engines for filling a partial table.

2. **Normalize the bad point.** Fix a point x at which E255 fails, and put

$$a = x \diamond x, \qquad q = a \diamond x = ((x \diamond x) \diamond x), \qquad p = x \backslash x.$$

Then the badness condition is

$$q \diamond x \neq x. \tag{B0}$$

Moreover, if $u \diamond x = x$, then $u = q$. Hence, under (B0), there is no solution to $u \diamond x = x$ at all. Equivalent bad conditions are

$$q \backslash x \neq x, \qquad ((x \diamond x) \diamond q) \diamond (x \diamond x) \neq x, \qquad (q \diamond x) \diamond q \neq p. \tag{B1}$$

These give three quick tests for any proposed table.

3. **Build the principal left orbit.** Define

$$c_0 = x, \qquad c_{k+1} = x \diamond c_k, \qquad d_k = c_k \diamond x.$$

Let d be the period of the L_x-orbit of x. The recurrence

$$c_{k-1} = c_k \diamond d_{k+1} \tag{O}$$

holds for all k modulo d, and

$$p = c_{d-1}, \qquad q = d_1 = c_{d-2}, \qquad p \diamond c_1 = q. \tag{O1}$$

In a bad point, $d \geq 4$. Also every orbit edge must avoid the two forbidden values

$$d_k = c_k \diamond x \notin \{x, c_{k-1}\}. \tag{O2}$$

For a labelled search, the first symmetry break is usually

$$x = 0, \qquad 0 \diamond 0 = 1, \qquad 0 \diamond 1 = 2, \qquad 0 \diamond 2 = 3,$$

with the rest of the L_x-cycle chosen according to the period d.

4. **Exploit the defective right column.** Since no element maps to x under $R_x(t) = t \diamond x$, the value x is missing from

$$\operatorname{im} R_x.$$

Finiteness therefore forces a nontrivial right collision. Choose either any collision or the first repetition in the right tail

$$\rho_0 = x, \qquad \rho_{n+1} = \rho_n \diamond x.$$

Thus there are $y \neq z$ and e with

$$y \diamond x = z \diamond x = e. \tag{C0}$$

Then (T) gives the rectangle

$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \backslash x, \tag{C1}$$

and the splitter condition

$$y \diamond e \neq z \diamond e. \tag{C2}$$

This is not a contradiction. It is the shape a genuine counterexample must repeatedly thread.

5. **Keep the fixed maps fixed-point-free.** Define

$$H(t) = (x \diamond t) \diamond x, \qquad F(t) = x \diamond (t \diamond x).$$

They satisfy

$$x \backslash t = t \diamond H(t), \qquad F(L_x(t)) = L_x(H(t)). \tag{F}$$

Thus F and H are conjugate finite self-maps. At a bad point neither has a fixed point. Equivalently, a counterexample must avoid

$$H(t) = t \qquad \text{and} \qquad F(t) = t$$

for every $t \in M$.

6. **Use the permutation-label viewpoint.** Because every left translation is bijective, write

$$y \diamond t = \sigma(y)(t), \qquad \sigma(y) \in \mathrm{Sym}(M).$$

The label map $y \mapsto \sigma(y)$ is injective. A right collision (C0) becomes

$$\sigma(y)(x) = \sigma(z)(x) = e, \qquad \sigma(y)(e) \neq \sigma(z)(e),$$

and the rectangle becomes

$$\sigma(\sigma(y)(e))(y) = e \backslash x = \sigma(\sigma(z)(e))(z). \tag{P}$$

This is often the cleanest way to search: construct an injective family of permutations whose pointwise multiplication table satisfies E677.

7. **Period four is the first test arena.** If $d = 4$, write

$$c_0 = x, \qquad c_1 = a, \qquad c_2 = q, \qquad c_3 = p, \qquad r = q \diamond x, \qquad s = p \diamond x.$$

Every bad period-four candidate must satisfy

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q, \tag{P4}$$

and

$$r \notin \{x, a, q\}. \tag{P4a}$$

Thus either $r = p$ or r leaves the principal orbit $\{x, a, q, p\}$.

In the delayed branch $r = p$, one also has

$$q \diamond a = q, \qquad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \qquad s \notin \{x, a, q, p\}. \tag{P4b}$$

In the external branch $r \notin \{x, a, q, p\}$, set

$$h = r \diamond q, \qquad t = q \diamond a, \qquad u = x \diamond r.$$

Useful forced cells are

$$q \diamond s = a, \qquad (p \diamond q) \diamond p = s, \qquad a \diamond h = r. \tag{P4c}$$

In particular, a four-element orbit-only table cannot be bad; the right tail must create external structure.

8. Period five gives compact internal collisions. If $d = 5$, write

$$c_0 = x, \qquad c_1 = a, \qquad c_2 = b, \qquad c_3 = q, \qquad c_4 = p,$$

and put

$$e = b \diamond x, \qquad g = q \diamond x, \qquad s = p \diamond x.$$

Then a bad period-five candidate satisfies

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q, \tag{P5}$$

$$g = a \diamond ((b \diamond a) \diamond b), \qquad q \diamond ((a \diamond q) \diamond a) = x, \qquad (a \diamond q) \diamond a = x \diamond (g \diamond q), \tag{P5a}$$

and the exclusions

$$e \notin \{x, a\}, \qquad g \notin \{x, b\}, \qquad s \notin \{x, q\}. \tag{P5b}$$

The orbit recurrence also gives

$$b \diamond g = a, \qquad q \diamond s = b, \qquad p \diamond a = q. \tag{P5c}$$

If the right tail stays inside the principal orbit through g, then $g \in \{a, q, p\}$. The three internal branches begin as follows:

$$\begin{aligned}
g = q &\implies b \diamond q = a, \quad (a \diamond q) \diamond a = (q \diamond q) \diamond q = q\backslash x, \quad a \diamond q \neq q \diamond q, \\
g = a &\implies b \diamond a = a, \quad (q \diamond a) \diamond q = b \diamond x = a\backslash x, \quad q \diamond a \neq b, \\
g = p &\implies b \diamond p = a, \quad q \diamond s = b, \quad s \notin \{x, q\}.
\end{aligned} \tag{P5d}$$

These branches are good places to apply the collision rectangle (C1) by hand.

9. For period six and beyond, use uniform ground constraints. If $d = 6$, write

$$c_0 = x, \quad c_1 = a, \quad c_2 = b, \quad c_3 = c, \quad c_4 = q, \quad c_5 = p,$$

and put

$$h = c \diamond x, \qquad g = q \diamond x.$$

Then

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q, \tag{P6}$$

$$h = a \diamond ((b \diamond a) \diamond b), \qquad g = b \diamond ((c \diamond b) \diamond c), \tag{P6a}$$

$$q \diamond ((a \diamond q) \diamond a) = x, \qquad (a \diamond q) \diamond a = x \diamond (g \diamond q), \tag{P6b}$$

with

$$b \diamond x \notin \{x, a\}, \quad h \notin \{x, b\}, \quad g \notin \{x, c\}, \quad p \diamond x \notin \{x, q\}. \tag{P6c}$$

For longer periods, repeatedly use

$$d_{k+2} = c_k \diamond \big((c_{k+1} \diamond c_k) \diamond c_{k+1}\big), \tag{U1}$$

$$x = d_k \diamond \big((c_k \diamond d_k) \diamond c_k\big), \tag{U2}$$

$$(c_k \diamond d_k) \diamond c_k = x \diamond \big((d_k \diamond x) \diamond d_k\big). \tag{U3}$$

10. **Use the known construction atlas as search pressure.** The public Equation 677 database should be read as empirical calibration, not as a proof. Its current finite examples all satisfy E255, and they cluster into a few recognizable positive families:

linear field magmas	$x \diamond y = (1-\alpha)x + \alpha y, \quad \Phi_{10}(\alpha) = 0,$
Steiner line magmas	$S(2,5,n)$ blocks, each a copy of the unique size-5 magma,
pencil magmas	one pivot with a small number of size-5 petals,
fiber bundles	product or twisted product structure with visible fibers,
piecewise-linear magmas	$x \diamond y = x + f(y-x)$ with f split by residues or cosets,
near-field style magmas	large L_0, R_0 cycles and non-linear field-extension behavior.

Most of these examples are right-cancellative, and hence cannot contain a bad point of the kind being sought here. The non-right-cancellative examples are still useful: they show how right columns can collapse while every left row remains bijective, but in every known case the collapses still thread the collision rectangles and preserve E255.

11. **Colored-slope lift branch.** One concrete counterexample route is now isolated enough to be worth exhausting by SAT. Let

$$M = \mathbb{F}_p \times \mathbb{F}_q, \qquad (y,i) \diamond (x,j) = \big(Ay + Bx,\, O_{[y:x]}(i,j)\big)$$

for $(y,x) \neq (0,0)$, with a separate operation on the zero fiber. For the primary target

$$p = 19, \quad q = 7, \quad A = 7, \quad B = 4, \quad \lambda_{\text{bad}} = 5,$$

the affine base $y * x = 7y + 4x$ satisfies E677 on $\mathbb{F}_{19}$, and

$$1 * 1 = 11, \qquad 11 * 1 = 5, \qquad 5 * 1 = 1.$$

Thus a fourth-power failure is forced if the color table at slope 5 is

$$O_5 = C_3, \qquad O_4(u,v) = u + 2v, \qquad O_{12}(u,v) = 6u + 2v,$$

and the remaining slope operations can be completed with row permutations satisfying all slope identities. The seed itself passes the whole $\lambda = 5$ identity:

$$O_5\big(i, O_{12}(j, O_4(O_5(i,j), i))\big) = j.$$

For $i = 0$ this is

$$O_5(0,j) = 3j, \quad O_4(3j,0) = 3j, \quad O_{12}(j,3j) = 5j, \quad O_5(0,5j) = j,$$

and for $i \neq 0$ it is

$$O_5(i,j) = j + i, \quad O_4(j+i,i) = j + 3i, \quad O_{12}(j, j+3i) = j + 6i, \quad O_5(i, j+6i) = j.$$

Moreover, for $s \neq 0$ the final C_3 row avoids s, so any full completion gives a bad point.

The obstruction is not the bad slope but the coupled side block. The slope maps force the simultaneous block

$$\{1, 2, 4, 6, 8, 12, 14, 16, 17\},$$

rather than independent repairs. In particular, from the fixed O_4 and O_{12} one gets the derived constraints

$$O_{11}(O_1(i,j),i) = 2(i-j),$$

so each column of O_{11} is a permutation and O_1 is determined by those columns, and

$$O_{18}\big(j, O_1(6i+2j,i)\big) = 4(j-i).$$

The naive local repair

$$O_6(u,v) = v, \qquad O_{11}(u,v) = u+v,$$

with O_{14} forced by $\lambda = 6$, and the analogous $\lambda = 17$ repair, satisfies the two neighboring equations but fails at the next compatibility equation $\lambda = 2$ once O_1 is forced. More generally, the affine/projection-side family

$$O_{11}(u,v) = au + bv + c, \qquad a, b \neq 0,$$

together with $O_6(u,v) = v$ and the forced O_{14} is UNSAT for all $6 \cdot 6 \cdot 7 = 252$ cases under the primary seed. Therefore a completion, if it exists, must make at least one of the O_{11}, O_6, O_{14} block genuinely non-affine or non-projection.

The search file is `explore_colored_magma.py`. It checks the slope equations, the full magma identity including the zero fiber, and the final E255 failure. Use the affine sweep to keep the dead repair family closed, then run deep branches over the remaining SAT instance. The most useful branch variable is O_{11}: the primary seed has a homogeneous color-scalar symmetry, so the branch $O_{11}(0,0) = 0$ plus the normalized nonzero branch $O_{11}(0,0) = 1$ covers the same quotient as all nonzero values. For machine exhaustion, split an entire column of O_{11} into permutations and shard it across workers; this is a partition of the remaining search, not an extra mathematical assumption.

12. **Use minimality as pressure.** A smallest counterexample cannot be commutative, right-cancellative, a quasigroup, or a loop, because the bad right column R_x misses x. The bad point cannot be idempotent, and its L_x-period is at least 4.

    As computational calibration only: the public database export has 841 known finite E677 magmas, all marked as satisfying E255. Sizes $2, 3, 4, 6, 8, 10$ have been proved empty for E677, while sizes $12, 14, 15$ remain open or only partially ruled out. Bad-witness searches have found no finite counterexample through order 9. At order 10, periods $4, 5, 6, 8, 9$ are closed in the existing search logs, leaving the broad period-7 and period-10 branches as the interesting order-10 frontier. These facts should guide effort, but a proposed counterexample still needs an independent table check.

13. **What counts as a finished counterexample.** A completed answer should provide a full Cayley table for M and verify:

    $$\text{E677 holds for all pairs } (x,y), \qquad \text{E255 fails at the chosen point } x.$$

    It should also report the data

    $$a = x \diamond x, \quad q = a \diamond x, \quad p = x \backslash x, \quad q \diamond x, \quad \text{the } L_x\text{-period } d,$$

and exhibit at least one right collision fiber satisfying the two displayed collision constraints. Also record the fingerprints: whether any pair-generated submagmas have size 5; whether the table is translation-invariant of the form $u \diamond v = u + f(v - u)$ after relabelling; whether the right-column collisions respect visible fibers; and the cycle structures of L_x and R_x at the bad point. If a branch is claimed impossible, the contradiction should use only the constraints above and left cancellation.

Final goal. Construct a finite table satisfying all of E677 and failing E255, or push one of the named branches to contradiction. A true counterexample must pass every local test above; a failed branch should explain exactly which forced equations collide.